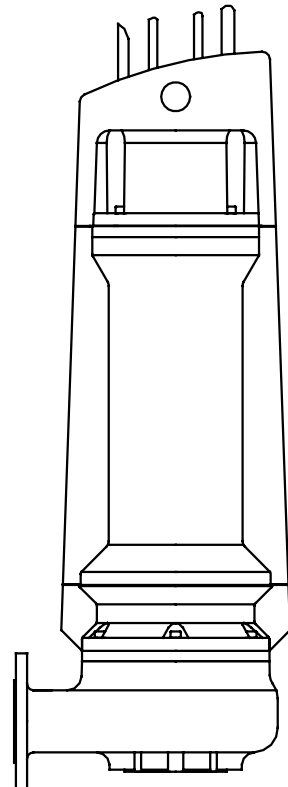
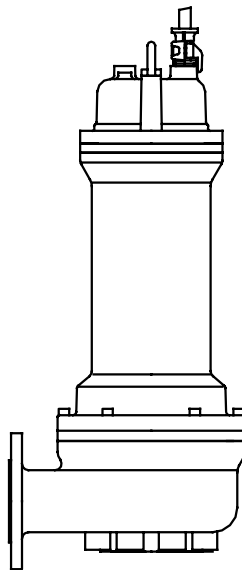
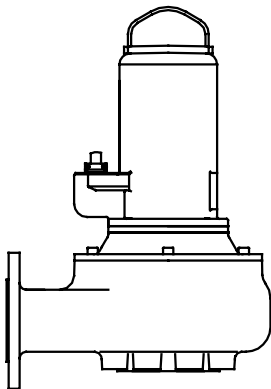


AP80 Vortex

- (GB) Installation and operating instructions
- (D) Montage- und Betriebsanleitung
- (F) Notice d'installation et d'entretien
- (I) Istruzioni di installazione e funzionamento
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Declaration of Conformity

We **GRUNDFOS** declare under our sole responsibility that the products **AP80 Vortex**, to which this declaration relates, are in conformity with the Council Directives on the approximation of the laws of the EEC Member States relating to

- Machinery (98/37/EEC).
Standard used: EN 292.
- Electromagnetic compatibility (89/336/EEC).
Standards used: EN 61 000-6-2 and EN 61 000-6-3.
- Electrical equipment designed for use within certain voltage limits (73/23/EEC).
Standards used: EN 60 335-1 and EN 60 335-2-41.

Konformitätserklärung

Wir **GRUNDFOS** erklären in alleiniger Verantwortung, daß die Produkte **AP80 Vortex**, auf die sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EG-Mitgliedstaaten übereinstimmen:

- Maschinen (98/37/EWG).
Norm, die verwendet wurde: EN 292.
- Elektromagnetische Verträglichkeit (89/336/EWG).
Normen, die verwendet wurden: EN 61 000-6-2 und EN 61 000-6-3.
- Elektrische Betriebsmittel zur Verwendung innerhalb bestimmter Spannungsgrenzen (73/23/EWG).
Normen, die verwendet wurden: EN 60 335-1 und EN 60 335-2-41.

Déclaration de Conformité

Nous **GRUNDFOS** déclarons sous notre seule responsabilité que les produits **AP80 Vortex** auxquels se réfère cette déclaration sont conformes aux Directives du Conseil concernant le rapprochement des législations des Etats membres CEE relatives à

- Machines (98/37/CEE).
Standard utilisé: EN 292.
- Compatibilité électromagnétique (89/336/CEE).
Standards utilisés: EN 61 000-6-2 et EN 61 000-6-3.
- Matériel électrique destiné à employer dans certaines limites de tension (73/23/CEE).
Standards utilisés: EN 60 335-1 et EN 60 335-2-41.

Dichiarazione di Conformità

Noi **GRUNDFOS** dichiariamo sotto la nostra esclusiva responsabilità che i prodotti **AP80 Vortex**, ai quali questa dichiarazione si riferisce, sono conformi alle Direttive del consiglio concernenti il ravvicinamento delle legislazioni degli Stati membri CEE relative a

- Macchine (98/37/CEE).
Standard usato: EN 292.
- Compatibilità elettromagnetica (89/336/CEE).
Standard usati: EN 61 000-6-2 e EN 61 000-6-3.
- Materiale elettrico destinato ad essere utilizzato entro certi limiti di tensione (73/23/CEE).
Standard usati: EN 60 335-1 e EN 60 335-2-41.

Declaración de Conformidad

Nosotros **GRUNDFOS** declaramos bajo nuestra única responsabilidad que los productos **AP80 Vortex** a los cuales se refiere esta declaración son conformes con las Directivas del Consejo relativas a la aproximación de las legislaciones de los Estados Miembros de la CEE sobre

- Máquinas (98/37/CEE).
Norma aplicada: EN 292.
- Compatibilidad electromagnética (89/336/CEE).
Normas aplicadas: EN 61 000-6-2 y EN 61 000-6-3.
- Material eléctrico destinado a utilizarse con determinados límites de tensión (73/23/CEE).
Normas aplicadas: EN 60 335-1 y EN 60 335-2-41.

Declaração de Conformidade

Nós **GRUNDFOS** declaramos sob nossa única responsabilidade que os produtos **AP80 Vortex** aos quais se refere esta declaração estão em conformidade com as Directivas do Conselho das Comunidades Europeias relativas à aproximação das legislações dos Estados Membros respeitantes à

- Máquinas (98/37/CEE).
Norma utilizada: EN 292.
- Compatibilidade electromagnética (89/336/CEE).
Normas utilizadas: EN 61 000-6-2 e EN 61 000-6-3.
- Material eléctrico destinado a ser utilizado dentro de certos limites de tensão (73/23/CEE).
Normas utilizadas: EN 60 335-1 e EN 60 335-2-41.

Δήλωση Συμμόρφωσης

Εμείς η **GRUNDFOS** δηλώνουμε με αποκλειστικά δική μας ευθύνη ότι τα προϊόντα **AP80 Vortex** συμμορφώνονται με την Οδηγία του Συμβουλίου επί της σύγκλισης των νόμων των Κρατών Μελών της Ευρωπαϊκής Ένωσης σε σχέση με τα

- Μηχανήματα (98/37/EEC).
Πρότυπο που χρησιμοποιήθηκε: EN 292.
- Ηλεκτρομαγνητική συμβατότητα (89/336/EEC).
Πρότυπα που χρησιμοποιήθηκαν: EN 61 000-6-2 και EN 61 000-6-3.
- Ηλεκτρικές συσκευές σχεδιασμένες για χρήση εντός ορισμένων ορίων ηλεκτρικής τάσης (73/23/EEC).
Πρότυπα που χρησιμοποιήθηκαν: EN 60 335-1 και EN 60 335-2-41.

Overeenkomstigheidsverklaring

Wij **GRUNDFOS** verklaren geheel onder eigen verantwoordelijkheid dat de producten **AP80 Vortex** waarop deze verklaring betrekking heeft in overeenstemming zijn met de Richtlijnen van de Raad inzake de onderlinge aanpassing van de wetgevingen van de Lid-Staten betreffende

- Machines (98/37/EEG).
Norm: EN 292.
- Elektromagnetische compatibiliteit (89/336/EEG).
Normen: EN 61 000-6-2 en EN 61 000-6-3.
- Elektrisch materiaal bestemd voor gebruik binnen bepaalde spanningsgrenzen (73/23/EEG).
Normen: EN 60 335-1 en EN 60 335-2-41.

Försäkran om överensstämmelse

Vi **GRUNDFOS** försäkrar under ansvar, att produkterna **AP80 Vortex**, som omfattas av denna försäkran, är i överensstämmelse med Rådets Direktiv om inbördes närmande till EU-medlemsstaternas lagstiftning, avseende

- Maskinell utrustning (98/37/EC).
Använd standard: EN 292.
- Elektromagnetisk kompatibilitet (89/336/EC).
Använda standarder: EN 61 000-6-2 och EN 61 000-6-3.
- Elektrisk material avsedd för användning inom vissa spänningsgränser (73/23/EC).
Använda standarder: EN 60 335-1 och EN 60 335-2-41.

Vastaavuusvakuutus

Me **GRUNDFOS** vakuutamme yksin vastuullisesti, että tuotteet **AP80 Vortex**, jota tämä vakuutus koskee, noudattavat direktiivejä jotka käsittelevät EY:n jäsenvaltioiden koneellisia laitteita koskevien lakien yhdenmukaisuutta seur.:

- Koneet (98/37/EY).
Käytetty standardi: EN 292.
- Elektromagneettinen vastaavuus (89/336/EY).
Käytetyt standardit: EN 61 000-6-2 ja EN 61 000-6-3.
- Määrätyt jänniterajojen puitteissa käytettävät sähköiset laitteet (73/23/EY).
Käytetyt standardit: EN 60 335-1 ja EN 60 335-2-41.

Overensstemmelseserklæring

Vi **GRUNDFOS** erklærer under ansvar, at produkterne **AP80 Vortex**, som denne erklæring omhandler, er i overensstemmelse med Rådets direktiver om indbyrdes tilnærmelse til EF medlemsstaternes lovgivning om

- Maskiner (98/37/EØF).
Anvendt standard: EN 292.
- Elektromagnetisk kompatibilitet (89/336/EØF).
Anvendte standarder: EN 61 000-6-2 og EN 61 000-6-3.
- Elektrisk materiel bestemt til anvendelse inden for visse spændingsgrænser (73/23/EØF).
Anvendte standarder: EN 60 335-1 og EN 60 335-2-41.

Bjerringbro, 15th August 2002



Kenth Hvid Nielsen
Technical Manager

AP80 Vortex

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Before beginning installation procedures, these installation and operating instructions should be studied carefully. The installation and operation should also be in accordance with local regulations and accepted codes of good practice.

1. General description

1.1 Applications

GRUNDFOS AP80 pumps with vortex impeller are designed for pumping:

- raw sewage,
- wastewater,
- wastewater with high concentration of fibrous material.

The pumps are designed to pump the above liquids in places such as:

- public buildings
- blocks of flats
- factories/industry
- garages
- underground car-parks
- car wash areas
- restaurants
- hotels

The compact design makes the pumps suitable for both submerged and dry permanent installation (dry installation applies only to pumps of 4.8 kW and up). In the case of submerged installation, the pumps can be installed by means of an auto-coupling guide rail system or they can be provided with a separate base stand for free-standing installation.

When fitted with a base stand and a flexible discharge hose, the pumps can also be used for temporary installations (as standby pumps) and can be moved from one installation to the other by means of the handle, lifting eye bolts or hole of the motor.

In the case of dry installation, the pump is supplied with a cooling jacket for internal or external cooling of the motor (dry installation applies only to pumps of 4.8 kW and up).

The AP80 pumps are designed with a free spherical impeller clearance of 80 mm.

1.1.1 Potentially explosive environments

Use the explosion-proof AP pump versions for applications involving the risk of explosion.

Note: The explosion classification of the pump is EEx de IIB T4 and **must** in each individual case be approved by the local authorities for use at the desired installation site.

Note: The separate motor starter/control box in standard version must not be installed in potentially explosive environments.

1.2 Operating conditions

1.2.1 pH Value

AP pumps in permanent installations can be used for pumping liquids with a pH value between 4 and 10.

1.2.2 Liquid temperature - submerged installation

Liquid temperature: 0°C to +40°C.

For short periods up to +60°C.

1.2.3 Liquid temperature - with cooling jacket

Liquid temperature: 0°C to +40°C.

For short periods up to +60°C.

1.2.4 Density of pumped liquid

Maximum density of pumped liquid: 1100 kg/m³.

1.2.5 Level of pumped liquid

In the case of submerged pump installation, the lowest stop level must always be above the top of the pump housing.

1.2.6 Operation

Maximum 20 starts per hour at intermittent operation.

Note: At continuous operation for more than one hour at a time, a cooling jacket must be installed if the motor is not completely submerged during operation.

1.3 Sound pressure level

The sound pressure level of the pump is lower than the limiting values stated in the EC Council Directive 98/37/EEC relating to machinery.

2. Safety



Pump installation in wells must be carried out by specially trained persons.

3. Transportation and storage

To ensure proper support during transportation, the pump is secured inside a sturdy carton when delivered from the factory. As a safety precaution, the carton must not be removed until the pump is to be installed.

Make sure that the pump cannot roll or fall over.

Always lift the pump by means of the handle, lifting eye bolts or hole, **never** by means of the motor cable or the hose/pipe.

For long periods of storage, the pump must be protected against moisture and heat.

After a long period of storage, the pump should be inspected before it is put into operation. Make sure that the impeller can rotate freely. Pay special attention to the condition of the shaft seals and the cable entry.

4. Installation

See mounting dimensions at the end of these instructions.

The loose nameplate supplied with the pump should be fixed at the installation site.

Prior to installation, check the oil level in the oil chamber, see section 7. *Maintenance and service*.

4.1 Submerged installation on auto-coupling

Pumps for permanent installation can be installed on a stationary auto-coupling and operated completely or partially submerged in the pumped liquid.

1. Drill mounting holes for guide rail bracket on the inside of the pit and fasten the guide rail bracket provisionally with two screws.
2. Place the auto-coupling base unit on the bottom of the pit. Use a plumb line to establish the correct positioning. Fasten with expansion bolts. If the bottom of the pit is uneven, the auto-coupling base unit must be supported so that it is level when being fastened.
3. Assemble the discharge line in accordance with the generally accepted procedures and without exposing the line to distortion or tension.
4. Insert the guide rails in the rings of the auto-coupling base unit and adjust the length of the rails accurately to the guide rail bracket.
5. Unscrew the provisionally fastened guide rail bracket, fit it on top of the guide rails and finally fasten it firmly to the pit wall.

Note: The guide rails must not have any axial play as this would cause noise during pump operation.

6. Clean out debris from the pit before lowering the pump into the pit.
7. Fit the auto-coupling half on to the discharge port of the pump. Then slide the guide bar of this coupling half between the guide rails and lower the pump into the pit by means of a chain secured to the eye bolts of the pump. When the pump reaches the auto-coupling base unit, the pump will automatically connect tightly.
8. Hang up the end of the chain on a suitable hook at the top of the pit.
9. Adjust the length of the motor cable by coiling it up on a relief fitting, so the cable is not damaged during operation. Fasten the relief fitting to a suitable bracket at the top of the pit. Make sure that the cables are not sharply bent or pinched.

Note: The end of the cable must not be submerged, as water may penetrate through the cable into the motor.

4.2 Free-standing submerged installation

1. Fit the separate base stand to the pump suction flange.
2. Fit a 90° elbow to the pump discharge port and connect the discharge pipe/hose.

If a hose is used, make sure that the hose does not buckle and that the inside diameter matches that of the discharge port.

3. Lower the pump into the liquid. It is recommended to position the pump on a solid foundation.

4.3 Dry installation

Applies only to pumps of 4.8 kW and up:

In dry installations, the pump is installed outside the pump sump. The pump motor is enclosed and watertight and will not be damaged if the installation site is flooded with water.

1. Fit the bracket and base stand to the pump.
2. Mark and drill mounting holes in the concrete floor.
3. Fasten the pump with expansion bolts.
4. Check that the pump is vertical/horizontal.
5. Connect the motor cable and the monitoring cable.

In order to facilitate service on the pump, isolating valves should be fitted on either side of the pump.

6. Fit the suction and discharge pipes and isolating valves, if used, and ensure that the pump is not stressed by the pipe-work.

4.4 Level controllers

The AP80 pumps can be supplied with a separate level controller: Type LC for one-pump installations and type LCD for two-pump installations.

The **LC** controller is fitted with two or three level switches. The third level switch, which is optional, is for high-level alarm.

The **LCD** controller is fitted with three or four level switches: one for common stop and two for start of pumps. The fourth level switch, which is optional, is for high-level alarm.

When installing the level switches, the following points should be observed:

1. To prevent air intake and vibrations in submerged pumps, the **stop level switch** must be fitted in such a way that the pump is stopped before the liquid level is lowered below the top of the pump housing.
2. The **start level switch** should be installed in such a way that the pump is started at the required level; however, the pump must always be started before the liquid level reaches the bottom inlet pipe in the pit.
3. The **high-level alarm switch**, if installed, should always be installed about 10 cm above the starting level switch; however, alarm must always be given before the liquid level reaches the inlet pipe to the pit.

4.5 Temperature sensors

Temperature sensors are built into the stator windings, and a contact in the sensors will open in case of overtemperature, i.e. 130°C.

Operating conditions:

Voltage [VAC]	Max. current [A]	cos φ
250	2.5	1.0
	1.6	0.6
500	0.75	1.0
	0.5	0.6

Non-explosion-proof pumps have only one set of temperature sensors.

Explosion-proof pumps have two sets of temperature sensors. The extra set of sensors will open at a temperature that is approx. 10°C higher than the opening temperature of standard sensors. The extra set of sensors provides additional protection against overtemperature in potentially explosive environments.

The temperature are connected to the monitoring cable, see section 5. *Electrical connection*, and must be connected to the safety circuit of the separate pump controller.

4.6 Moisture sensors

Applies only to pumps of 4.8 kW and up:

Pumps for submerged installation are available with or without moisture sensors.

Pumps with moisture sensors have one sensor in the oil chamber and another in the chamber below the top cover.

Pumps for dry installation are always supplied with one moisture sensor only, which is in the oil chamber.

A moisture sensor consists of two probes between which the resistance (conductivity) is measured. In case of moisture ingress, the resistance will change depending on the amount and conductivity of the entering liquid.

In order to enable motor cutout in case of a leak, the pump controller must include a tripping unit for each sensor. This unit must maintain an electrically separated AC 24 voltage across the sensor and should have an adjustable sensitivity of 0 to 100 kΩ.

The sensitivity of the tripping unit should be set to approx. 50 kΩ, depending on the conductivity of the pumped liquid.

The moisture sensors are connected to the monitoring cable, see section 5. *Electrical connection*, and the tripping units of these sensors must be connected to the safety circuit of the separate pump controller.

5. Electrical connection

The electrical connection should be carried out in accordance with local regulations.

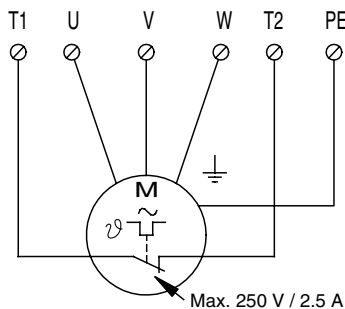
The operating voltage and frequency are marked on the name-plate. Voltage tolerance: $\pm 10\%$ of the voltage stated on the name-plate ($\pm 5\%$ for explosion-proof pumps). Make sure that the motor is suitable for the electricity supply available at the installation site.



The pump must be connected to an external mains switch.

AP80 pumps up to and including 2.6 kW are for direct-on-line starting only, see fig. 1.

Fig. 1

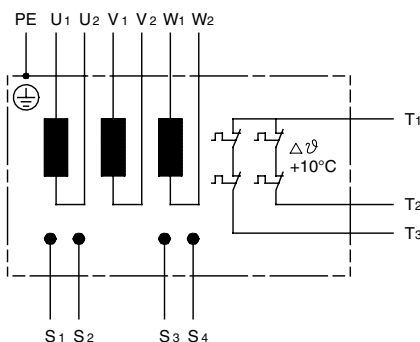


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Other pump types are prepared for star-delta starting, i.e. both ends of the motor windings are accessible through the motor cable, see fig. 2.

- T1 and T3 are connected to the standard set of temperature sensors.
- T1 and T2 are connected to the extra set of temperature sensors in explosion-proof pumps.
- S1 and S2 are connected to the moisture sensor in the oil chamber.
- S3 and S4 are connected to the moisture sensor in the chamber below the top cover.
- Pumps with two moisture sensors can be connected to a common alarm relay (parallel connection) or to two alarm relays.

Fig. 2



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5.1 Motor protection

All pumps are supplied with 10 metres of cable and a free cable end. The pumps must be connected to a separate pump controller with motor starter.

The stator windings incorporate temperature sensors which cut out the motor in case of overtemperature.

Moisture sensors have been incorporated in the oil chamber between the motor and the pump housing as well as at the terminal board under the top cover. The moisture sensors will cut out the motor in case of water penetration (applies only to pumps of 4.8 kW and up).

The temperature and moisture sensors are connected to the monitoring cable. The temperature sensors and the tripping units of the moisture sensors must be connected to the safety circuit of the separate pump controller.

Note: The separate motor starter/control box in standard version must not be installed in potentially explosive environments.

6. Start-up

Proceed as follows:

1. Check the oil level in the oil chamber.
2. Remove the fuses and check whether the impeller can rotate freely.
3. Check whether the monitoring units, if used, are operating satisfactorily.
4. Check whether the system has been filled with liquid and vented.
5. Make sure that the pump is submerged in the liquid.
6. Open the isolating valves, if fitted.
7. Check the setting of the level switches.
8. Start the pump.
9. In dry installations, the cooling jacket fitted to the pump must be vented separately.

Note: The pump may, however, be started for a very short period without being submerged for checking of the direction of rotation.

6.1 Checking of direction of rotation

The direction of rotation should be checked every time the pump is connected to a new installation.

Check the direction of rotation as follows:

1. Start the pump and check the quantity of water or the discharge pressure.
2. Stop the pump and interchange two of the phases to the motor.
3. Start the pump and check the quantity of water or the discharge pressure.
4. Stop the pump.
5. Compare the results taken under points 1 and 3, and the connection which gives the larger quantity of water or the higher pressure is the correct direction of rotation.

7. Maintenance and service



Before starting work on the pump, make sure that the electricity supply has been switched off and that it cannot be accidentally switched on. Furthermore, all rotating parts must have stopped moving.

Maintenance and service must be carried out by specially trained persons.

Before carrying out maintenance and service, it must be ensured that the pump has been thoroughly flushed with clean water. Rinse the pump parts in water after dismantling.



When unscrewing the inspection screw of the oil chamber, please note that pressure may have built up in the chamber. Do not remove the screw until the pressure has been fully relieved.

Pumps running normal operation should be inspected at least once a year. If the pumped liquid is very muddy or sandy or if the pump is operating continuously, the pump should be inspected every 1000 operating hours.

When the pump is new or after replacement of the shaft seals, check the oil level after one week of operation.

For long and trouble-free operation of the pump, the following points should be checked regularly:

- **Power consumption**
- **Oil level and oil condition**
The oil becomes greyish white like milk if it contains water. This may be the result of a defective shaft seal. The oil should be replaced after 3000 operating hours.
Note: Used oil must be disposed of in accordance with local regulations.
The oil chamber contains 1.7 litres Shell Ondina 15 oil.
- **Cable entry**
Make sure that the cable entry is watertight and that the cables are not bent sharply and/or pinched.

- **Pump parts**

Check the impeller, pump housing, etc. for possible wear. Replace defective parts.

- **Ball bearings**

Check the shaft for noisy or heavy operation (turn the shaft by hand). Replace defective ball bearings.

A general overhaul of the pump is usually required in case of defective ball bearings or poor motor function. This work must be carried out by an authorized GRUNDFOS service workshop.

7.1 Service kits

The service kit for AP80 pumps up to and including AP80.80.26 includes: 1 primary shaft seal, 1 secondary shaft seal, 1 O-ring kit, 1 cable entry. The service kit for AP80 pumps from AP80.80(100).48 up to and including AP80.80(100).107 includes: 1 primary shaft seal, 1 secondary shaft seal, 1 O-ring kit, 1 cable entry.	The service kit for AP80 pumps from AP80.80(100).160 up to and including AP80.80(100).250 includes: 1 primary shaft seal, 1 secondary shaft seal, 1 O-ring kit, 1 rubber bush for cable entry, 1 oil filling screw.
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Part numbers for service kits can be found in the tables at the end of these instructions.

7.2 Contaminated pumps

Note: If a pump has been used for a liquid which is injurious to health or toxic, the pump will be classified as contaminated.

If GRUNDFOS is requested to service the pump, GRUNDFOS must be contacted with details about the pumped liquid, etc. *before* the pump is returned for service. Otherwise GRUNDFOS can refuse to accept the pump for service.

Possible costs of returning the pump are paid by the customer.

However, any application for service (no matter to whom it may be made) must include details about the pumped liquid if the pump has been used for liquids which are injurious to health or toxic.

8. Fault finding chart



Make sure that the electricity supply has been switched off and that all rotating parts have stopped moving before attempting to diagnose any fault.

GB

Fault	Cause	Remedy
1. Motor does not start. Fuses blow or motor starter trips out immediately. Caution: Do not start again!	a) Supply failure; short-circuit; earth-leakage fault in cable or motor winding.	Have the cable and motor checked and repaired by a qualified electrician.
	b) Fuses blow due to use of wrong type of fuse.	Install fuses of the correct type.
	c) Impeller blocked by impurities.	Clean the impeller.
	d) Level switch out of adjustment or defective.	Check the level switch.
2. Pump operates, but motor starter trips out after a short while.	a) Low setting of thermal relay in motor starter.	Set the relay in accordance with the specifications on the pump nameplate.
	b) Increased current consumption due to large voltage drop.	Measure the voltage between two motor phases. Tolerance: $\pm 10\%$ ($\pm 5\%$ for explosion-proof pumps).
	c) Impeller blocked by impurities. Increased current consumption in all three phases.	Clean the impeller.
3. Pump operates at below-standard performance and power consumption.	a) Impeller blocked by impurities.	Clean the impeller.
	b) Wrong direction of rotation.	Check the direction of rotation and possibly interchange two phases. See section 6.1 <i>Checking of direction of rotation</i> .
4. Pump operates, but gives no liquid.	a) Discharge valve closed or blocked.	Check the discharge valve and open and/or clean.
	b) Non-return valve blocked.	Clean the non-return valve.
	c) Air in pump.	Vent the pump.
5. Pump cycling between start and stop.	a) Level switch out of adjustment or defective.	Check the level switch.

9. Disposal

Disposal of this product or parts of it must be carried out according to the following guidelines:

1. Use the local public or private waste collection service.
2. In case such waste collection service does not exist or cannot handle the materials used in the product, please deliver the product or any hazardous materials from it to your nearest GRUNDFOS company or service workshop.

Subject to alterations.

AP80.80.10 - 26.V.(Ex)

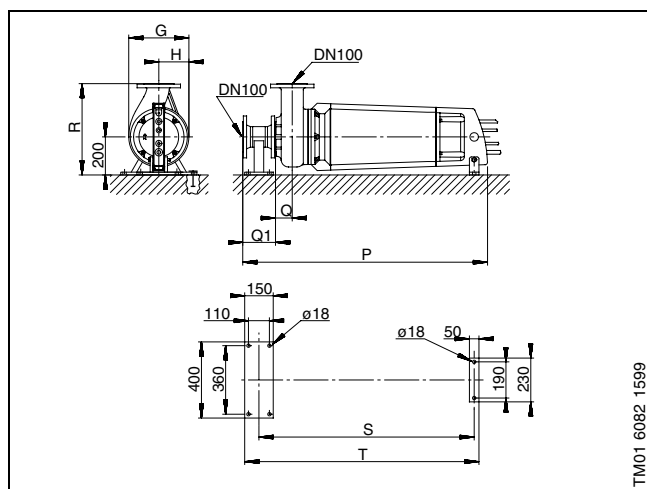
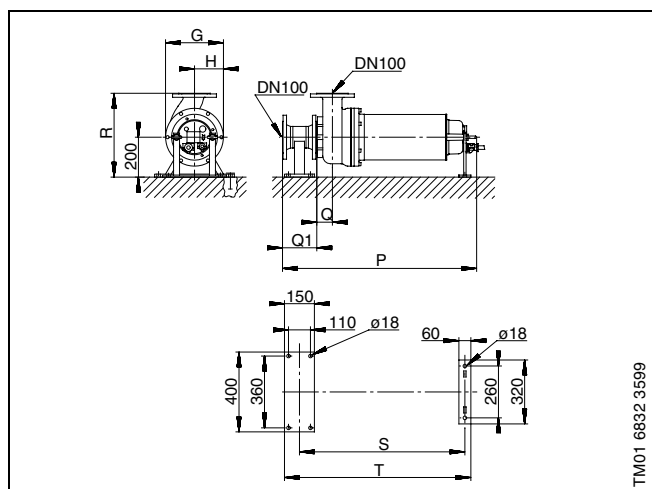
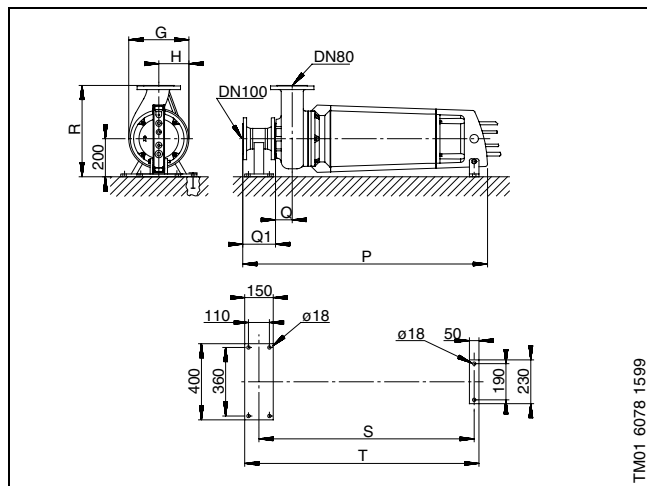
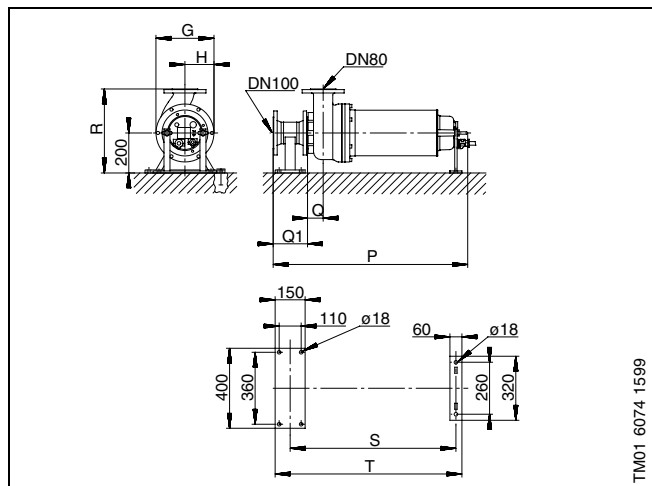
AP80	Standard version	Explosion-proof version
AP80.80.10	96 00 53 75	96 00 53 76
AP80.80.13		
AP80.80.20		
AP80.80.26		

GB	Pump type	Standard pump (with moisture sensors)	Without moisture sensors	Dry installation, version D	Dry installation, horizontal version DH
D	Pumpentyp	Standardpumpe (mit Feuchtigkeitsfühlern)	Ohne Feuchtigkeitsfühler	Trockenaufstellung, Version D	Trockenaufstellung, horizontal, Version DH
F	Type de pompe	Pompe standard (avec capteurs d'humidité)	Sans capteurs d'humidité	Installation non immergée, version D	Installation non immergée, horizontale, version DH
I	Tipo di pompa	Pompa standard (con sensori di umidità)	Senza sensori di umidità	Installazione a secco, versione D	Installazione a secco, orizzontale, versione DH
E	Tipo de bomba	Bomba estándar (con sensores de humedad)	Sin sensores de humedad	Instalación en seco, versión D	Instalación en seco, horizontal, versión DH
P	Modelo da bomba	Bomba standard (com sensores de humidade)	Sem sensores de humidade	Instalação em seco, versão D	Instalação em seco, horizontal, versão DH
GR	Τύπος αντλίας	Κανονική εκδοχή (αισθητήρες υγρασίας)	Χωρίς αισθητήρες υγρασίας	Εγκατάσταση εν ξηρώ, εκδοχή D	Οριζόντια εγκατάσταση εν ξηρώ, εκδοχή DH
NL	Pomptype	Standaard pomp (met vochtdetectie opnemers)	Zonder vochtdetectie opnemers	Droge installatie, versie D	Droge installatie, horizontaal, versie DH
S	Pumptyp	Standardpump (med fuktsensorer)	Utan fuktsensorer	Torruppställd installation, version D	Torruppställd installation horisontal, version DH
SF	Pumppumalli	Vakiopumppu (vuoto-antureilla)	Ilman vuoto-antureita	Kuiva-asennus, rakenne D	Kuiva-asennus vaaka, rakenne DH
DK	Pumpetype	Standardpumpe (med fugtighedsfølere)	Uden fugtighedsfølere	Tørtopstillet installation, version D	Tørtopstillet installation horisontal, version DH

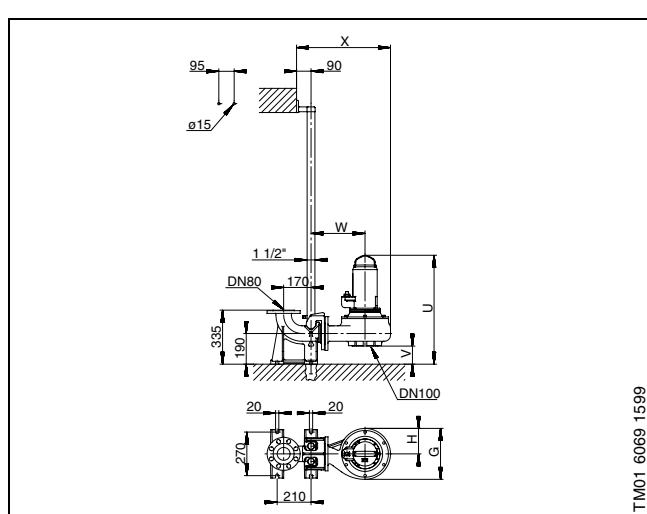
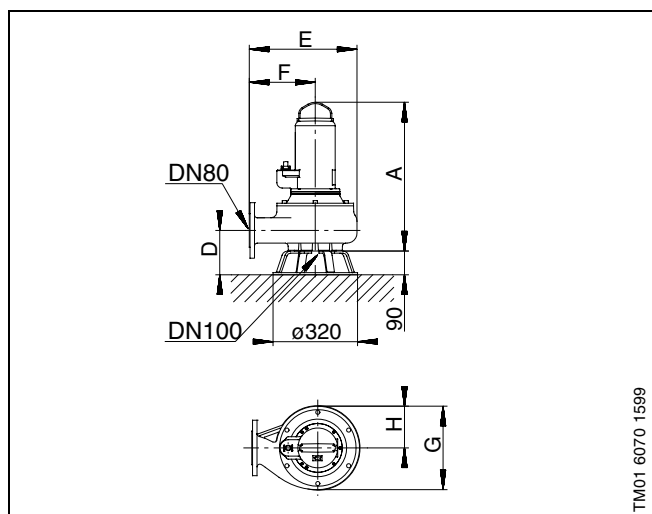


Pump type	Standard pump (with moisture sensors)	Without moisture sensors	Dry installation, version D	Dry installation, horizontal version DH
AP80.80.48.V	96 00 53 78	96 00 53 77	96 00 53 79	96 00 53 79
AP80.80.65.V				
AP80.80.92.V				
AP80.80.107.V				
AP80.80.160.V	96 00 53 83	96 00 53 82	96 00 53 85	96 00 53 85
AP80.80.190.V			96 00 53 86	96 00 53 86
AP80.80.250.V	96 00 53 84			
AP80.100.48.V	96 00 53 78	96 00 53 77	96 00 53 79	96 00 53 79
AP80.100.65.V				
AP80.100.92.V				
AP80.100.107.V				
AP80.100.160.V	96 00 53 83	96 00 53 82	96 00 53 85	96 00 53 85
AP80.100.190.V			96 00 53 86	96 00 53 86
AP80.100.250.V	96 00 53 84			
AP80.80.48.V.Ex	96 00 53 80	96 00 53 77	96 00 53 81	96 00 53 81
AP80.80.65.V.Ex				
AP80.80.92.V.Ex				
AP80.80.107.V.Ex				
AP80.80.160.V.Ex	96 00 53 87	96 00 53 87	96 00 53 88	96 00 53 88
AP80.80.190.V.Ex				
AP80.80.250.V.Ex				
AP80.100.48.V.Ex	96 00 53 80	96 00 53 77	96 00 53 81	96 00 53 81
AP80.100.65.V.Ex				
AP80.100.92.V.Ex				
AP80.100.107.V.Ex				
AP80.100.160.V.Ex	96 00 53 87	96 00 53 87	96 00 53 88	96 00 53 88
AP80.100.190.V.Ex				
AP80.100.250.V.Ex				

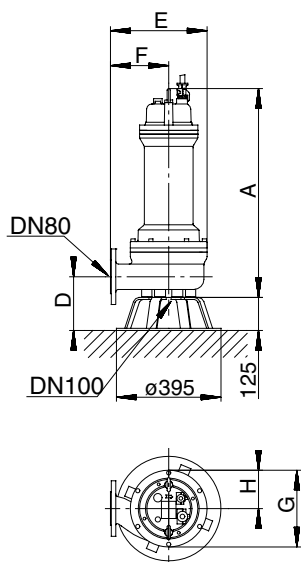
Installation drawings



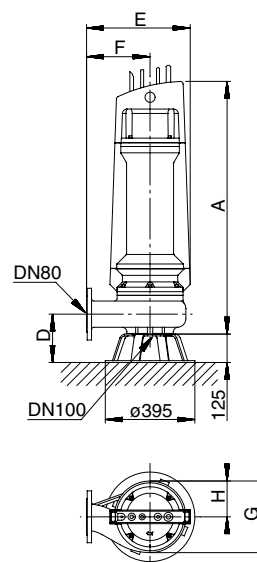
Pump type	G	H	P	Q	Q1	R	S	T
AP80.80.48 - 65	290	145	903	78	172	420	759	864
AP80.80.92 - 107	290	145	973	78	172	420	829	934
AP80.80.160 - 190	316	158	1196	78	172	450	1041	1141
AP80.80.250	316	158	1286	78	172	450	1131	1231
AP80.100.48 - 65	290	145	903	88	172	440	759	864
AP80.100.92 - 107	290	145	973	88	172	440	829	934
AP80.100.160 - 190	316	158	1196	88	172	480	1041	1141
AP80.100.250	316	158	1286	88	172	480	1131	1231



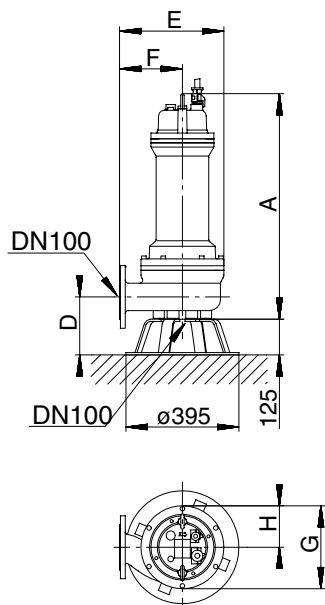
Pump type	A	D	E	F	G	H	U	V	W	X
AP80.80.10 - 13	516	168	380	220	290	145	628	112	305	540
AP80.80.20 - 26	563	168	410	250	316	158	675	112	335	583



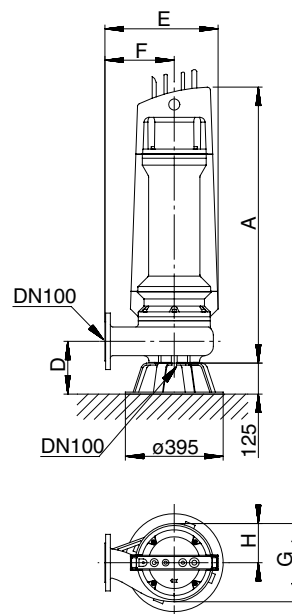
TM01 6072 1599



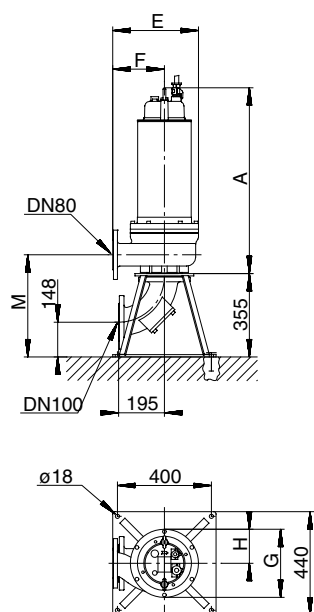
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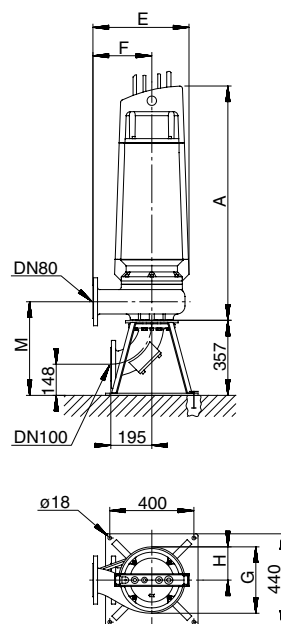
TM01 6834 3399



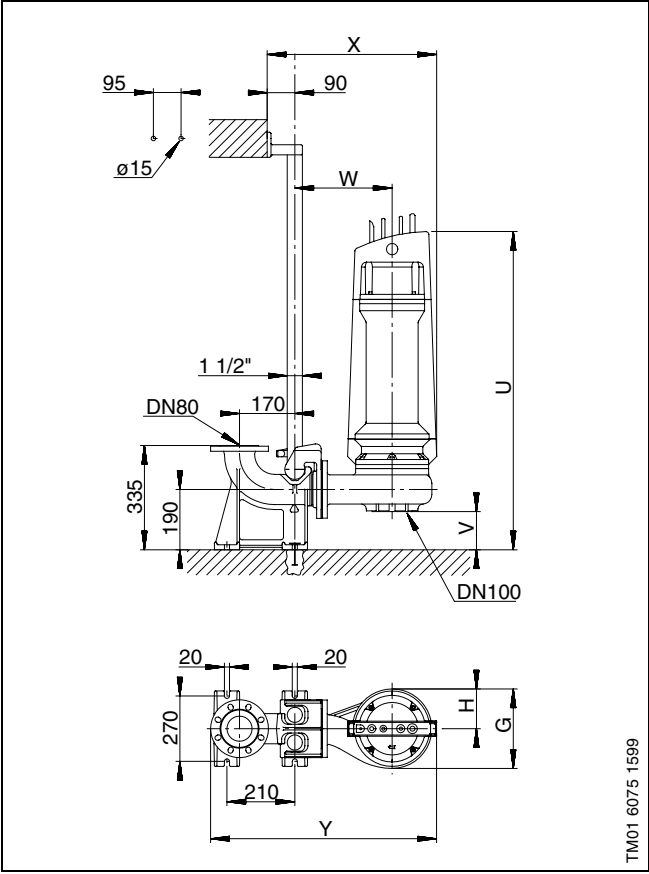
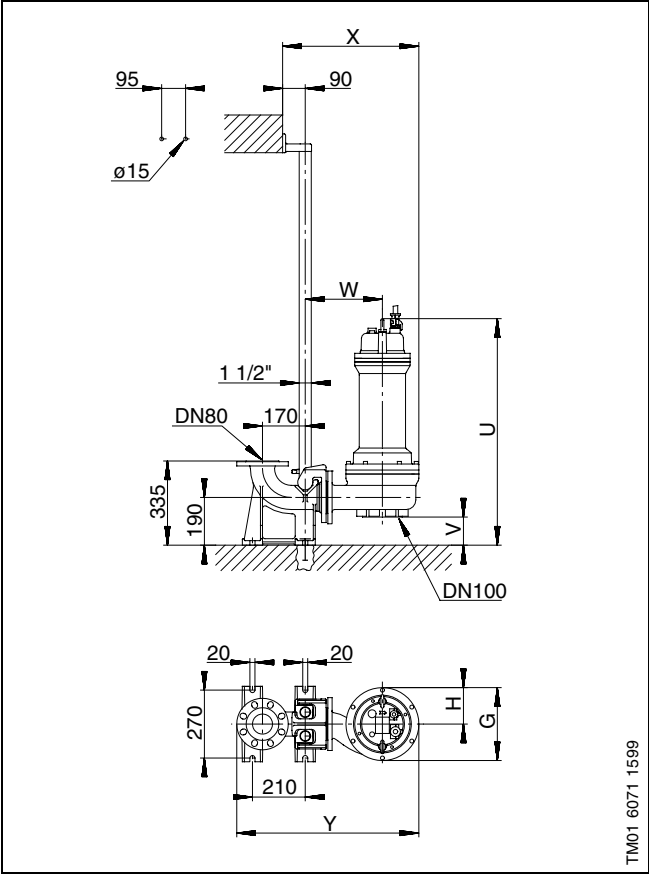
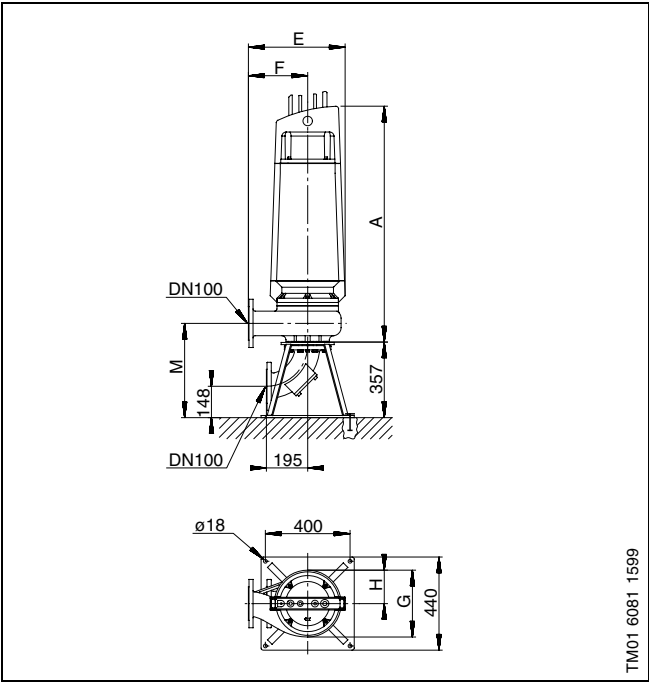
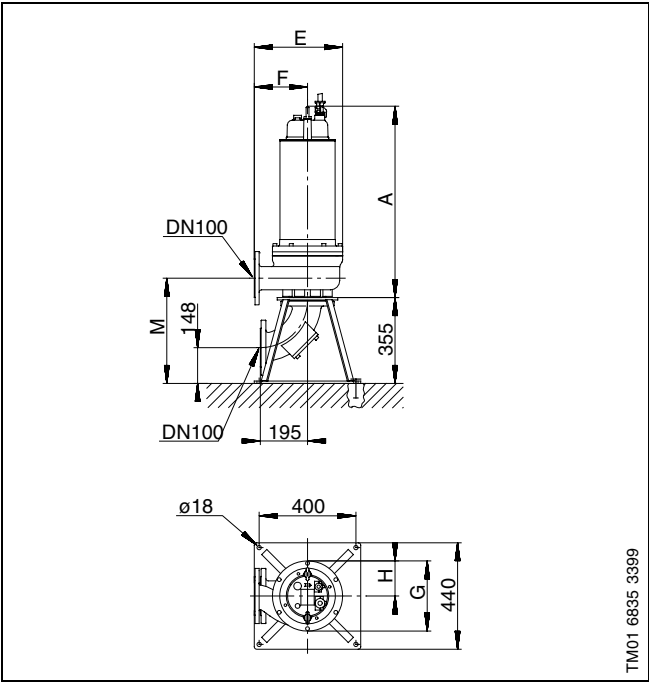
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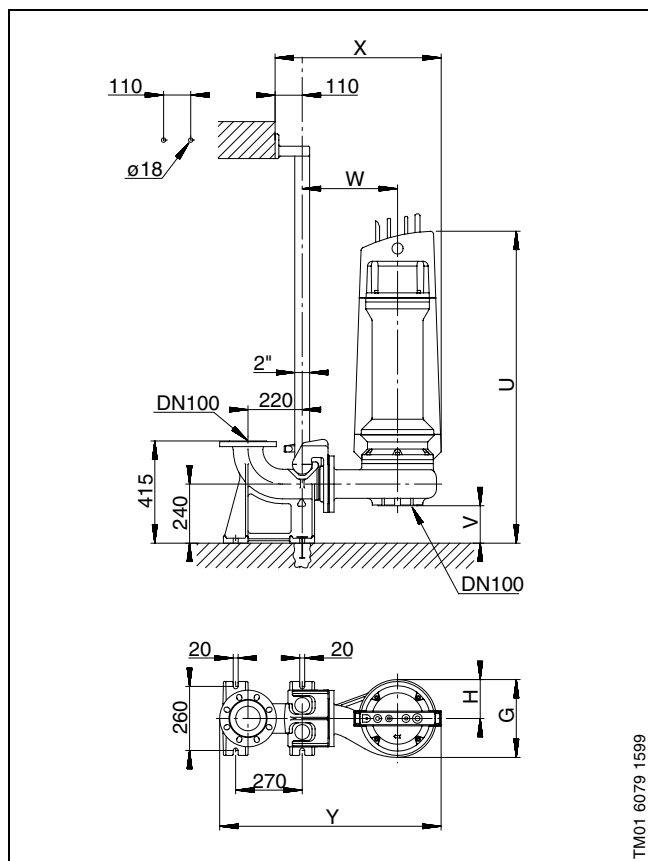
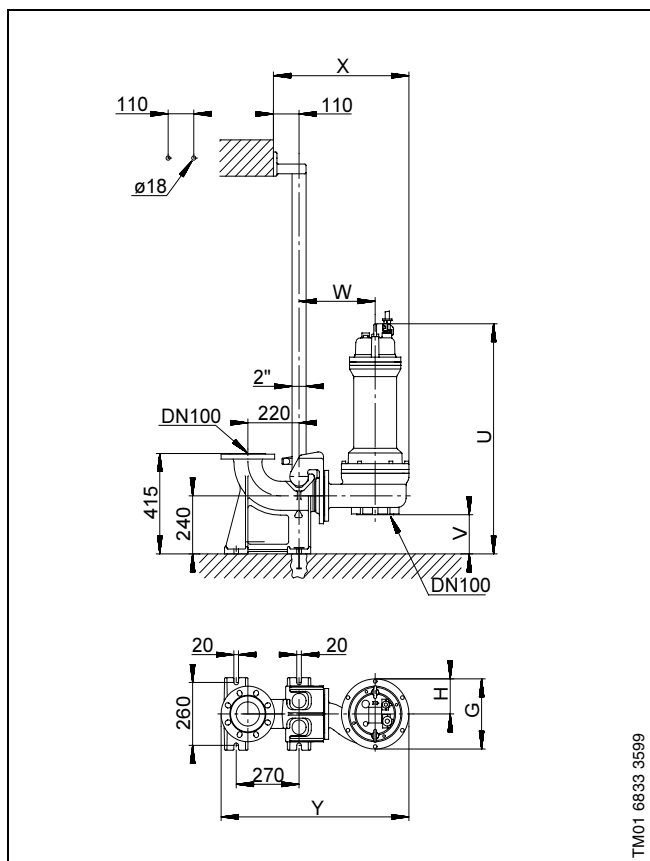


TM01 6073 1599



TM01 6077 1599





Pump type	A	D	E	F	G	H	M	U	V	W	X	Y
AP80.80.48 - 65	719	203	365	220	290	145	435	831	112	307	542	725
AP80.80.92 - 107	789	203	365	220	290	145	435	901	112	307	542	725
AP80.80.160 - 190	1024	203	427	250	316	158	435	1136	112	335	602	785
AP80.80.250	1114	203	427	250	316	158	435	1226	112	335	602	785
AP80.100.48 - 65	719	213	385	240	290	145	445	871	152	347	602	827
AP80.100.92 - 107	789	213	385	240	290	145	445	941	152	347	602	827
AP80.100.160 - 190	1027	213	457	280	316	158	445	1176	152	387	674	899
AP80.100.250	1117	213	457	280	316	158	445	1266	152	387	674	899

Denmark

GRUNDFOS DK A/S
Poul Due Jensens Vej 7A
DK-8850 Bjerringbro
Tlf.: +45-87 50 50 50
Telefax: +45-87 50 51 51
E-mail: info_GDK@grundfos.com
www.grundfos.com/DK

Argentina

Bombas GRUNDFOS de Argentina S.A.
Ruta Panamericana km. 37.500 Lote 34A
1619 - Garin
Pcia. de Buenos Aires
Phone: +54-3327 414 444
Telefax: +54-3327 411 111

Australia

GRUNDFOS Pumps Pty. Ltd.
P.O. Box 2040
Regency Park
South Australia 5942
Phone: +61-8-8461-4611
Telefax: +61-8-8340 0155

Austria

GRUNDFOS Pumpen Vertrieb Ges.m.b.H.
Grundfosstraße 2
A-5082 Grödig/Salzburg
Tel.: +43-6246-883-0
Telefax: +43-6246-883-30

Belgium

N.V. GRUNDFOS Bellux S.A.
Boomsesteenweg 81-83
B-2630 Aartselaar
Tél.: +32-3-870 7300
Télécopie: +32-3-870 7301

Brazil

GRUNDFOS do Brasil Ltda.
Rua Tomazina 106
CEP 83325 - 040
Pinhais - PR
Phone: +55-41 668 3555
Telefax: +55-41 668 3554

Canada

GRUNDFOS Canada Inc.
2941 Brighton Road
Oakville, Ontario
L6H 6C9
Phone: +1-905 829 9533
Telefax: +1-905 829 9512

China

GRUNDFOS Pumps (Shanghai) Co. Ltd.
22 Floor, Xin Hua Lian Building
755-775 Huai Hai Rd, (M)
Shanghai 200020
PRC
Phone: +86-512-67 61 11 80
Telefax: +86-512-67 61 81 67

Czech Republic

GRUNDFOS s.r.o.
Cajkovského 21
779 00 Olomouc
Phone: +420-585-716 111
Telefax: +420-585-438 906

Finland

OY GRUNDFOS Pumput AB
Mestarintie 11
Piispankylä
FIN-01730 Vantaa (Helsinki)
Phone: +358-9 878 9150
Telefax: +358-9 878 91550

France

Pompes GRUNDFOS Distribution S.A.
Parc d'Activités de Chesnes
57, rue de Malacombe
F-38290 St. Quentin Fallavier (Lyon)
Tél.: +33-4 74 82 15 15
Télécopie: +33-4 74 94 10 51

Germany

GRUNDFOS GMBH
Schlüterstr. 33
40699 Erkrath
Tel.: +49-(0) 211 929 69-0
Telefax: +49-(0) 211 929 69-3799
e-mail: infoservice@grundfos.de
Service in Deutschland:
e-mail: kundendienst@grundfos.de

Greece

GRUNDFOS Hellas A.E.B.E.
20th km. Athinon-Markopoulou Av.
P.O. Box 71
GR-19002 Peania
Phone: +0030-210-66 83 400
Telefax: +0030-210-66 46 273

Hong Kong

GRUNDFOS Pumps (Hong Kong) Ltd.
Unit 1, Ground floor
Siu Wai Industrial Centre
29-33 Wing Hong Street &
68 King Lam Street, Cheung Sha Wan
Kowloon
Phone: +852-27861706/27861741
Telefax: +852-27858664

Hungary

GRUNDFOS Hungária Kft.
Park u. 8
H-2045 Törökbálint,
Phone: +36-23 511 110
Telefax: +36-23 511 111

India

GRUNDFOS Pumps India Private Limited
Flat A, Ground Floor
61/62 Chamiers Aptmt
Chamiers Road
Chennai 600 028
Phone: +91-44 432 3487
Telefax: +91-44 432 3489

Indonesia

PT GRUNDFOS Pompa
Jl. Rawa Sumur III, Blok III/CC-1
Kawasan Industri, Pulogadung
Jakarta 13930
Phone: +62-21-460 6909
Telefax: +62-21-460 6910/460 6901

Ireland

GRUNDFOS (Ireland) Ltd.
Unit 34, Stillorgan Industrial Park
Blackrock
County Dublin
Phone: +353-1-2954926
Telefax: +353-1-2954739

Italy

GRUNDFOS Pompe Italia S.r.l.
Via Gran Sasso 4
I-20060 Truccazzano (Milano)
Tel.: +39-02-95838112
Telefax: +39-02-95309290/95838461

Japan

GRUNDFOS Pumps K.K.
1-2-3, Shin Miyakoda
Hamamatsu City
Shizuoka pref. 431-21
Phone: +81-53-428 4760
Telefax: +81-53-484 1014

Korea

GRUNDFOS Pumps Korea Ltd.
6th Floor, Aju Building 679-5
Yeoksam-dong, Kangnam-ku, 135-916
Seoul Korea
Phone: +82-2-5317 600
Telefax: +82-2-5633 725

Malaysia

GRUNDFOS Pumps Sdn. Bhd.
7 Jalan Peguam U1/25
Glenmarie Industrial Park
40150 Shah Alam
Selangor
Phone: +60-3-5569 2922
Telefax: +60-3-5569 2866

Mexico

Bombas GRUNDFOS de Mexico S.A. de C.V.
Boulevard TLC No. 15
Parque Industrial Stiva Aeropuerto
Apodaca, N.L. 66600
Mexico
Phone: +52-81-8144 4000
Telefax: +52-81-8144 4010

Netherlands

GRUNDFOS Nederland B.V.
Postbus 104
NL-1380 AC Weesp
Tel.: +31-294-492 211
Telefax: +31-294-492244/492299

New Zealand

GRUNDFOS Pumps NZ Ltd.
17 Beatrice Tinsley Crescent
North Harbour Industrial Estate
Albany, Auckland
Phone: +64-9-415 3240
Telefax: +64-9-415 3250

Norway

GRUNDFOS Pumper A/S
Strømsveien 344
Postboks 235, Leirdal
N-1011 Oslo
Tlf.: +47-22 90 47 00
Telefax: +47-22 32 21 50

Poland

GRUNDFOS Pompy Sp. z o.o.
ul. Klonowa 23
Baranowo k. Poznania
PL-62-081 Przemierowo
Phone: +48-61-650 13 00
Telefax: +48-61-650 13 50

Portugal

Bombas GRUNDFOS Portugal, S.A.
Rua Calvet de Magalhães, 241
Apartado 1079
P-2770-153 Paço de Arcos
Tel.: +351-21-440 76 00
Telefax: +351-21-440 76 90

Russia

OOO GRUNDFOS
Shkolnaya 39
RUS-109544 Moscow
Phone: +7-095 564 88 00, +7-095 737 30 00
Telefax: +7-095 564 88 11, +7-095 737 75 36
e-mail: grundfos.moscow@grundfos.com

Singapore

GRUNDFOS (Singapore) Pte. Ltd.
24 Tuas West Road
Jurong Town
Singapore 638381
Phone: +65-6865 1222
Telefax: +65-6861 8402

Spain

Bombas GRUNDFOS España S.A.
Camino de la Fuentequilla, s/n
E-28110 Algete (Madrid)
Tel.: +34-91-848 8800
Telefax: +34-91-628 0465

Sweden

GRUNDFOS AB
Box 63, Angeredsvinkeln 9
S-424 22 Angered
Tel.: +46-771-32 23 00
Telefax: +46-31 331 94 60

Switzerland

GRUNDFOS Pumpen AG
Bruggacherstrasse 10
CH-8117 Fällanden/ZH
Tel.: +41-1-806 8111
Telefax: +41-1-806 8115

Taiwan

GRUNDFOS Pumps (Taiwan) Ltd.
14, Min-Yu Road
Tunglo Industrial Park
Tunglo, Miao-Li County
Taiwan, R.O.C.
Phone: +886-37-98 05 57
Telefax: +886-37-98 05 70

Thailand

GRUNDFOS (Thailand) Ltd.
947/168 Moo 12, Bangna-Trad Rd., K.M. 3,
Bangna, Phrakonong
Bangkok 10260
Phone: +66-2-744 1785 ... 91
Telefax: +66-2-744 1775 ... 6

Turkey

GRUNDFOS POMPA SAN. ve TIC. LTD. STI
Bulgurlu Caddesi no. 32
TR-81190 Üsküdar İstanbul
Phone: +90 - 216-4280 306
Telefax: +90 - 216-3279 988

United Arab Emirates

GRUNDFOS Gulf Distribution
P.O. Box 16768
Jebel Ali Free Zone
Dubai
Phone: +971-4- 8815 166
Telefax: +971-4-8815 136

United Kingdom

GRUNDFOS Pumps Ltd.
Grovebury Road
Leighton Buzzard/Beds. LU7 8TL
Phone: +44-1525-850000
Telefax: +44-1525-850011

U.S.A.

GRUNDFOS Pumps Corporation
17100 West 118th Terrace
Olathe, Kansas 66061
Phone: +1-913-227-3400
Telefax: +1-913-227-3500

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